Brain MRI Module 1
Dataset & Modeling Guide

Ground-truth columns, age-aware vs age-independent ablations, split strategy, and class-imbalance handling

1. Final Ground-Truth Sheet: Column Meaning

- ANON_PatientID - Main de-identified patient identifier. Use this to connect each spreadsheet row to the corresponding anonymized patient data. It is a linkage field, not a model input.
- ANON_StudyInstanceUID - Anonymous study-level identifier. Keep it as a second linkage/QC key so the correct MRI study can be verified for each patient. It is not a model input.
- Classification - Main target label for Module 1. Values are NORMAL or ABNORMAL. This is what the classifier must predict.
- Age - Patient age at scan. Use as an additional model feature only in the age-aware experiment. Do not use age as a model input in the age-independent experiment.
- Diagnosis - Reference/helper field describing the radiological finding (for example chronic stroke, acute stroke, tiny lacune, hemorrhage). Do not feed this to the classifier.
- Significance - Reference/helper field. The value non_significant marks mild age-related cases used to define the two ablation strategies. Do not feed this directly to the model.

Model leakage rule: ANON_PatientID, ANON_StudyInstanceUID, Diagnosis, and Significance are metadata/helper columns and must not be used as predictive inputs.

2. Dataset / Ablation 1 - Age-Aware Classification

Label setup

- Total patients: 856.
- Normal: 400.
- Abnormal: 456.
- Among the 400 Normal cases, 41 are marked non_significant and represent mild age-related findings that are considered clinically insignificant in the appropriate age context.
- The 41 non_significant cases are a subset of the Normal class, not a third target class.

Why age must be used

In this dataset, some mild findings are labeled according to clinical age context. Therefore, the image alone may not contain all information used to assign the label. Age should be supplied as a separate patient-level feature.

Recommended age-aware model flow

- Coronal T2-FLAIR slices
- DINOv2 slice-level feature extraction
- Position-aware attention pooling across slices
- Patient-level MRI embedding
- Concatenate normalized age
- Classifier / MLP
- NORMAL vs ABNORMAL

Important: Do not pass age into DINOv2. Concatenate normalized age after the patient-level MRI representation has been created.

Age normalization: Use training-set statistics only, e.g. normalized_age = (age - training_mean) / training_std. Do not calculate the mean or standard deviation using validation or test patients.

3. Handling the 41 non_significant Cases in Ablation 1

- These 41 cases are especially important because they are where the age-aware strategy is most clinically relevant.
- Use most of them in training, but retain a small representative subset in validation so the model's handling of these cases can be evaluated.
- Recommended split for these 41 cases: approximately 34 for training and 7 for validation.
- Keep the split strictly patient-level. All slices from one patient must stay entirely within one split.
- Do not split slices independently across training and validation.
- If a separate final test cohort is maintained, keep it fully unseen by the training and validation process.

Practical rule: stratify the patient-level split and explicitly check that the validation set contains a few non_significant cases.

4. Dataset / Ablation 2 - Age-Independent Classification

- In this version, all 41 non_significant age-related cases are relabeled as ABNORMAL.
- Normal: 359.
- Abnormal: 497.
- Total patients remains 856.
- Age is not provided to the classifier in this experiment.

Recommended age-independent model flow

- Coronal T2-FLAIR slices
- DINOv2 slice-level feature extraction
- Position-aware attention pooling
- Patient-level MRI embedding
- Classifier / MLP
- NORMAL vs ABNORMAL

Interpretation of the comparison: Because both the labeling policy and the model inputs differ between the two experiments, do not claim that any performance difference is caused only by adding age. Frame the comparison as age-aware clinical classification vs age-independent abnormality classification.

5. Class Imbalance Handling for Ablation 2

Ablation 2 has 359 Normal vs 497 Abnormal cases. This is a moderate imbalance, not an extreme one. Do not force an artificial 50:50 dataset unless experiments show it is necessary.

Recommended practices

- Stratified patient-level splitting - Preserve approximately the same Normal/Abnormal ratio across train and validation/test splits.
- Training-only data augmentation - Apply augmentation only after the split and only to the training set.
- Suitable mild augmentations - Small rotations, translations, mild zoom/scale, limited intensity/contrast variation, and carefully chosen noise.
- Avoid aggressive augmentation - Do not use transformations that may distort or remove subtle abnormalities such as tiny lacunes.
- Class-weighted loss - Consider weighted cross-entropy/BCE so errors on the minority Normal class receive slightly higher importance.

- Weighted sampling - Optionally use a WeightedRandomSampler so Normal patients are sampled more often during training.
- Avoid over-correction - Do not combine very aggressive oversampling with very large class weights unless justified by experiments.
- Validation/test sets - Do not augment or duplicate samples merely to make these sets balanced. Keep them representative of the true data distribution.

6. Metrics to Report

- Accuracy
- Sensitivity / Recall
- Specificity
- Precision
- F1-score
- ROC-AUC
- Confusion matrix
- Balanced accuracy (recommended, especially for the age-independent dataset)

Do not rely only on accuracy, because the larger Abnormal class can make accuracy look acceptable even when the model performs poorly on Normal cases.

7. Data Augmentation and Split Order

Correct order: patient-level split first, then augment training data only.

1. Start with the 856 patients.
2. Create patient-level Train / Validation / Test partitions.
3. Keep every slice from a patient inside the same partition.
4. Apply augmentation only to the training partition.
5. Never augment the validation or test sets.

Why this matters: Augmenting before splitting can place near-duplicate images from the same patient in different splits, creating data leakage.

8. Summary of the Two Experiments

Item	Ablation 1: Age-Aware	Ablation 2: Age-Independent
Total patients	856	856
Normal	400	359
Abnormal	456	497
non_significant cases	41 retained as Normal	41 relabeled Abnormal
Age supplied to classifier?	Yes	No
MRI sequence	Coronal T2-FLAIR	Coronal T2-FLAIR
Image encoder	DINOv2	DINOv2
Patient aggregation	Position-aware attention	Position-aware attention
Target	NORMAL / ABNORMAL	NORMAL / ABNORMAL
Imbalance handling	Mostly stratification; no aggressive balancing needed	Stratification + augmentation + class weighting/weighted sampling if useful

9. What Is Actually Fed to the Model

Ablation 1 - Age-Aware

- Model inputs: MRI + normalized Age.
- Target: Classification.

- Not model inputs: ANON_PatientID, ANON_StudyInstanceUID, Diagnosis, Significance.

Ablation 2 - Age-Independent

- Model input: MRI only.
- Target: Classification.
- Not model inputs: Age, ANON_PatientID, ANON_StudyInstanceUID, Diagnosis, Significance.

Use of helper columns: Diagnosis and Significance should be retained for error analysis. For example, after inference, the team can check whether failures are concentrated in tiny lacunes, chronic stroke, hemorrhage, or age-related minimal changes.

10. Team Handoff Checklist

- Match spreadsheet rows to the anonymized DICOM data using ANON_PatientID, with ANON_StudyInstanceUID available as an additional QC key.
- Use Classification as the only target label.
- For Ablation 1, feed normalized Age after the patient-level MRI embedding.
- For Ablation 2, do not feed Age.
- Keep the split patient-level; never split slices from the same patient across partitions.
- Use approximately 34 of the 41 non_significant cases for training and 7 for validation in the age-aware setup.
- Apply augmentation only to the training set.
- For Ablation 2, address the 359 vs 497 imbalance using stratification first, then class weights / weighted sampling / moderate augmentation as needed.
- Evaluate with sensitivity, specificity, F1, ROC-AUC, confusion matrix, and balanced accuracy in addition to accuracy.
- Use Diagnosis and Significance only for reference and downstream error analysis.